

MOC

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1 Replicability Intervention: Canonical Phase Cells and the Revised Safe-Anchor Program

This section incorporates the methodological intervention that replaced arbitrarily precise phase tracking by a deterministic, reproducible finite-resolution state.

The intervention changes the mathematical formulation of the upper-bound problem.

The previous post-checkpoint analysis established that the reduced profile

$$y = (m, \mu_2/2, \mu_4/10)$$

does not determine the descendant state: numerically indistinguishable reduced profiles can correspond to different transport couplings and produce different descendants. Thus the phase fiber is dynamically relevant. The previous formulation therefore enlarged the state from

$$V_m(y)$$

to

$$V_m(y, \Psi) \quad \text{or} \quad V_m(y, \xi).$$

The new intervention is more restrictive.

The aim is no longer to recover the exact phase. Instead, one seeks a finite-resolution phase label together with a deterministic reconstruction rule.

The central question becomes:

How many robust phase bits per dyadic scale are sufficient for safe continuation?

Throughout, statements are labelled

$$[\text{PROVEN}], \quad [\text{NUMERICAL}], \quad [\text{OPEN}].$$

1.1 Why exact phase tracking is the wrong state description

Let

$$\mathcal{P}_m(A, B)$$

denote the transport polytope at scale m , and let

$$F_m : \mathcal{P}_m(A, B) \longrightarrow \mathbb{R}^3$$

be the reduced profile map.

The post-checkpoint experiments gave numerical pairs

$$\pi_1, \pi_2 \in \mathcal{P}_m(A, B)$$

with

$$F_m \pi_1 \approx F_m \pi_2$$

but

$$m(\pi_1) \not\approx_m (\pi_2).$$

Thus the fiber

$$\ker F_m$$

contains dynamically relevant information.

A naive response would be to retain the complete phase

$$\Psi_m(\pi)$$

with increasingly high numerical precision.

The replicability intervention rejects this requirement.

There are two reasons.

First, the full fresh phase space grows with the renormalization scale:

$$\dim K_m = m^2.$$

Second, exact real-valued phase coordinates can silently carry an unbounded amount of discrete information.

Hence an upper-bound theorem based on arbitrarily precise phase tracking would leave unresolved whether the dynamics is genuinely finite-information.

The correct computational state should therefore be robust under small numerical changes.

1.2 Canonicalization before quantization

The first requirement is to eliminate arbitrary selection inside a degenerate transport fiber.

For a feasible reduced target y , define

$$\mathcal{F}_m(y) = \{\pi \in \mathcal{P}_m(A, B) : F_m \pi = y\}.$$

A generic zero-objective linear program does not define a continuous selection

$$y \mapsto \pi.$$

Numerically, arbitrarily small changes of y can cause the solver to jump between different basic feasible solutions.

This was observed directly in finite-difference experiments: the corresponding descendant-support Jacobian changed by order one when the finite-difference scale was reduced.

Hence:

[NUMERICAL] an arbitrary LP vertex selector is not a stable phase section.

A canonical section is instead a deterministic map

$$C_m : y \mapsto C_m(y) \in \mathcal{F}_m(y).$$

A natural smooth choice is the maximum-relative-entropy section

$$C_m^{\text{ME}}(y) = \arg \max_{\pi \in \mathcal{F}_m(y)} \left[- \sum_j \pi_j \log \frac{\pi_j}{r_j} \right],$$

where $r_j > 0$ is a fixed reference measure.

On a relative-interior feasible fiber the objective is strictly concave, so the maximizing coupling is unique.

Thus the relation

$$y \longmapsto \{\pi\}$$

is replaced by a canonical function

$$y \longmapsto C_m^{\text{ME}}(y).$$

1.3 Interior targets are essential

The critical reduced target is

$$T_s = \left(\frac{7}{9}, \frac{2}{3}, \frac{3}{5} \right).$$

On the audited $N = 129$ state,

$$\delta_{\text{prof}}(T_s) = -0.0121013203478 \dots,$$

so the exact critical target lies outside the current reachable profile region.

Consequently, attempting to solve the maximum-entropy moment equations exactly at T_s drives the dual parameter toward the boundary rather than toward a finite interior solution.

This explains the previously observed apparently pathological maximum-entropy cold starts. By contrast, at a genuine interior safe center with

$$\delta_{\text{prof}}(y_0) \gtrsim 5 \times 10^{-4},$$

the same maximum-entropy solve converges normally.

One audited interior point was approximately

$$y_0 = (0.7650653, 0.6605157, 0.6056618),$$

with finite dual parameter

$$\lambda_0 \approx (487.98, 127.58, -44.10).$$

Thus:

[NUMERICAL] canonical MaxEnt continuation is well behaved on the safe interior, not at the infeasible critical target.

This distinction is structurally important.

The invariant-tube problem concerns a robust neighborhood of the critical fingerprint, not exact equality with the fingerprint at every finite scale.

1.4 Stable finite-difference response after canonicalization

Let

$$h_i(y) = h_{\mathcal{R}_{(C_m^{\text{ME}}(y))}}(z_i)$$

denote child support functions along selected Farkas directions.

A finite-difference Jacobian was computed using dyadic perturbations

$$\Delta_1 = 2^{-14}, \quad \Delta_2 = 2^{-17}.$$

After canonicalization, the two Jacobians satisfy

$$\varepsilon_J = \frac{\|J(\Delta_1) - J(\Delta_2)\|_F}{\|J(\Delta_2)\|_F} \approx 1.6 \times 10^{-3}.$$

This is in sharp contrast with the noncanonical LP section, where the Jacobian changed discontinuously under scale refinement.

Hence:

[NUMERICAL] canonicalization converts the phase response into a reproducible finite-resolution operator.

1.5 Effective phase-information spectrum

Let

$$J_{\text{raw}} \in \mathbb{R}^{8 \times 3}$$

be the finite-resolution support-response operator obtained by probing three canonical parent control coordinates and eight child Farkas directions.

Let

$$Z = \begin{pmatrix} z_1^\top \\ \vdots \\ z_8^\top \end{pmatrix},$$

and define the orthonormalized three-dimensional facet embedding

$$Q = Z(Z^\top Z)^{-1/2}, \quad Q^\top Q = I_3.$$

The profile-like effective response is

$$J_{\text{eff}} = Q^\top J_{\text{raw}}.$$

On the first audited canonical state,

$$\sigma(J_{\text{eff}}) \approx (1.972, 1.008, 0.249).$$

The same computation at the two finite-difference scales gave

$$(1.972, 1.008, 0.249)$$

and

$$(1.970, 1.008, 0.249),$$

respectively.

Thus the spectrum is stable at the level required for the present diagnostic.

The corresponding unweighted positive-log expansion is

$$R_{\text{bits}}^{\text{eff}} = \sum_{j=1}^3 \log_2^+ \sigma_j \approx 0.99.$$

Almost the entire value comes from one expanding direction:

$$\log_2(1.972) \approx 0.98,$$

while

$$\log_2(1.008) \approx 0.01, \quad \log_2^+(0.249) = 0.$$

Thus the first canonical audit suggests the effective structure

one expanding mode, one nearly marginal mode, one strongly contracting mode.

This is not yet a theorem and is not yet known to be state-independent.

1.6 Shape response is not negligible

The eight-dimensional support response is not entirely contained in the three-dimensional profile-like image.

Define

$$R_{\text{shape}} = (I - QQ^\top)J_{\text{raw}}.$$

Numerically,

$$\frac{\|R_{\text{shape}}\|_F}{\|J_{\text{raw}}\|_F} \approx 0.235.$$

Therefore the value

$$R_{\text{bits}}^{\text{eff}} \approx 0.99$$

must not yet be interpreted as the total phase-information rate.

It prices only the effective profile-like channel selected by Q .

A complete rate analysis must retain the full

$$8 \times 3$$

operator or prove that the shape component is uniformly controlled.

1.7 From exact phase to finite phase cells

Let

$$\xi_m(\pi) \in \mathbb{R}^d$$

be a finite-dimensional phase descriptor.

Introduce the dyadic quantizer

$$Q_b(x) = 2^{-b} \text{round}(2^b x).$$

The new phase state is not

$$\xi_m(\pi)$$

itself, but its finite label

$$q_m = Q_b(\xi_m(\pi)).$$

The corresponding phase cell is

$$\mathcal{C}_m(q) = \{\pi : Q_b(\xi_m(\pi)) = q\}.$$

Thus the previous state description

$$(y, \Psi)$$

is replaced schematically by

$$(y, q), \quad q \in 2^{-b}\mathbb{Z}^d.$$

The intended canonical physical representative is

$$\widehat{\pi}_m = C_m(y, q),$$

where C_m is one fixed deterministic reconstruction rule.

1.8 Quantization margin

A finite quantized label is useful only if it is stable under numerical perturbation.

Define the coordinatewise quantization margin

$$\text{qmargin}_b(\xi) = \min_j \text{dist} \left(\xi_j, 2^{-b} \left(\mathbb{Z} + \frac{1}{2} \right) \right).$$

A robust phase state satisfies

$$\text{qmargin}_b(\xi) \geq \gamma 2^{-b}$$

for some

$$\gamma > 0.$$

Then every perturbation satisfying

$$\|\delta\xi\|_\infty < \gamma 2^{-b}$$

leaves the quantized phase label unchanged.

Hence reproducibility becomes an explicit margin condition rather than an informal demand for more floating-point precision.

1.9 Quantization-loss lemma

Suppose the descendant safety margin obeys a local Lipschitz estimate

$$|\rho_{2m}(\pi) - \rho_{2m}(\tilde{\pi})| \leq L_\xi \|\xi_m(\pi) - \xi_m(\tilde{\pi})\|.$$

For coordinatewise dyadic quantization,

$$\|\xi - Q_b(\xi)\| \leq c_d 2^{-b},$$

where c_d depends only on the chosen finite phase dimension and norm.

Therefore

$$\rho_{2m}(\hat{\pi}) \geq \rho_{2m}(\pi) - L_\xi c_d 2^{-b}.$$

Consequently, if

$$\rho_{2m}(\pi) \geq \rho_0 + L_\xi c_d 2^{-b},$$

then its canonical quantized realization still satisfies

$$\rho_{2m}(\hat{\pi}) \geq \rho_0.$$

Thus finite precision can be deliberately traded for reproducibility without destroying safety, provided a uniform slack exists.

This is the basic deterministic analogue of an error-correcting margin.

1.10 The revised invariant tube

The old phase-augmented tube had the schematic form

$$\mathfrak{R}_m = \{(y, \xi) : \|y - T_s\|_\infty \leq \varepsilon_0, \delta_{\text{prof}}(y) \geq \rho_0, \xi \in \Xi\}.$$

The replicability intervention replaces Ξ by a finite collection of robust phase cells.

Define

$$\mathfrak{K}_m^{(b)} = \left\{ (y, q) : \begin{array}{l} \|y - T_s\|_\infty \leq \varepsilon_0, \\ \delta_{\text{prof}}(y) \geq \rho_0, \\ q \in \mathcal{Q}_m^{\text{safe}} \end{array} \right\},$$

where

$$\mathcal{Q}_m^{\text{safe}} \subseteq 2^{-b}\mathbb{Z}^d$$

is the finite set of admissible robust phase labels.

The scale-doubling problem is now

$$(y_m, q_m) \in \mathfrak{K}_m^{(b)} \implies \exists \text{ canonical admissible lift } \hat{\pi}_m$$

such that

$$(y_{2m}, q_{2m}) \in \mathfrak{K}_{2m}^{(b)}.$$

The phase need not be tracked exactly.

Only its finite robust cell must propagate.

1.11 Finite-state viability interpretation

If the number of phase bits b and the descriptor dimension d are independent of m , then the phase component of the scale dynamics has only finitely many relevant states.

Schematically,

$$(y_m, q_m) \longmapsto (y_{2m}, q_{2m}).$$

The Bellman problem therefore changes from a growing-dimensional continuous control problem into a finite-resolution viability problem.

Instead of

$$V_m(y, \Psi)$$

on an m^2 -dimensional fresh-phase sector, one seeks

$$V_m^{(b)}(y, q).$$

The theorem-level objective is not convergence of one greedy orbit.

It is closure of the viable family:

$$\forall (y, q) \in \mathfrak{K}_m^{(b)} \quad \exists \hat{\pi}_m \quad (y', q') \in \mathfrak{K}_{2m}^{(b)}.$$

1.12 Replicable Safe-Anchored Lemma

The central missing statement can now be formulated sharply.

Conjecture 1 (Replicable Safe-Anchored Lemma). *There exist constants*

$$b < \infty, \quad d < \infty, \quad \varepsilon_0 > 0, \quad \rho_0 > 0, \quad \gamma > 0, \quad C_{\text{bill}} < \infty,$$

independent of the dyadic scale m , together with

$$\xi_m : \text{admissible couplings} \longrightarrow \mathbb{R}^d,$$

a dyadic quantizer Q_b , and a deterministic canonical reconstruction rule C_m , such that the following holds.

For every

$$\Theta_m \in \mathfrak{K}_m^{(b)}$$

there exists a prefix-compatible canonical lift

$$\widehat{\pi}_m = C_m(F_m \pi_m, Q_b(\xi_m(\pi_m)))$$

satisfying

$$\|F_m \widehat{\pi}_m - T_s\|_\infty \leq \varepsilon_0,$$

$$\delta_{\text{prof}}(F_m \widehat{\pi}_m) \geq \rho_0,$$

$$\text{qmargin}_b(\xi_m(\widehat{\pi}_m)) \geq \gamma 2^{-b},$$

and whose child again lies in the same scale-independent replicable tube:

$$\Theta_{2m} =_m(\widehat{\pi}_m) \in \mathfrak{K}_{2m}^{(b)}.$$

Moreover, the same lift admits a valid causal inventory continuation with

$$\Delta \mathcal{B}_m \leq C_{\text{bill}}.$$

The last condition is essential.

Reduced-profile viability alone does not imply the inventory upper bound.

1.13 Inventory bridge remains a separate theorem obligation

Let

$$\mathcal{K}_n$$

denote the set of root laws of valid depth- n zero-repair trees.

The exact common-reserve recycling statement remains:

if one common reserve law R satisfies

$$B_h * R \in \mathcal{K}_n \quad \forall |h| = n$$

for every depth- n leaf law B_h , then

$$B * R \in \mathcal{K}_{2n}.$$

The root inventory increment is exactly

$$\mathbb{E}R.$$

Thus a uniform bound

$$\mathbb{E}R \leq C_{\text{bill}}$$

would give

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C_{\text{bill}}.$$

The replicability intervention does not remove this obligation.

Rather, it changes the candidate safe-anchor object on which the common reserve must be constructed.

The desired bridge is therefore

$$\text{replicable phase cell} \longrightarrow \text{prefix-compatible safe descendant} \longrightarrow \text{common reserve of } O(1) \text{ mean.}$$

1.14 Consequence for the inventory asymptotics

Assume the Replicable Safe-Anchor Lemma, including the inventory clause.

Then iterating the doubling estimate gives

$$\mathcal{B}_{2^k n_0} \leq \mathcal{B}_{n_0} + k C_{\text{bill}}.$$

Since

$$k = \log_2 \frac{n}{n_0},$$

one obtains

$$\boxed{\mathcal{B}_n = O(\log n).}$$

The rigorous lower bound already gives

$$\boxed{\mathcal{B}_n = \Omega(\log n).}$$

Therefore:

$$\boxed{\text{Replicable Safe-Anchor Lemma} + \text{uniform inventory bridge} \implies \mathcal{B}_n = \Theta(\log n).}$$

Equivalently, if the inventory increment is included directly in the statement of the Replicable Safe-Anchor Lemma as above, then proving that single combined lemma closes the asymptotic problem.

1.15 Phase-bit sufficiency

The new formulation suggests an information-theoretic quantity.

Let

$$b_m^*$$

be the smallest number of phase bits for which one can construct a replicable safe anchor at scale m .

The decisive asymptotic question is

$$\boxed{\sup_m b_m^* < \infty ?}$$

If yes, the amount of dynamically relevant phase information per dyadic renormalization step is uniformly bounded.

More generally, for a d -dimensional phase channel with linearized effective operator J_m , define the local expansion proxy

$$\boxed{R_m^{\text{phase}} = \sum_{j=1}^d \log_2^+ \sigma_j(J_m).}$$

For the first canonical audit,

$$R_m^{\text{phase}} \approx 0.99$$

in the profile-like three-dimensional channel.

This suggests approximately one expanding phase bit per dyadic scale on that state.

However:

$$\boxed{[\text{OPEN}] \quad R_m^{\text{phase}} \text{ is not yet known to be uniformly bounded in } m \text{ or over the viable tube.}}$$

1.16 Weighted phase rate

The raw singular values depend on the input and output norms.

The theorem-relevant operator should therefore eventually be normalized by the available parent quantization radius and child safety slack.

Let

$$W_{\text{in}}, \quad W_{\text{out}}$$

be the corresponding positive metric operators.

Define

$$\tilde{J}_m = W_{\text{out}} J_m W_{\text{in}}^{-1}.$$

Then the normalized phase-rate proxy is

$$R_{m,\text{bits}} = \sum_j \log_2^+ \sigma_j(\tilde{J}_m).$$

The natural uniform information conjecture is

$$\sup_m R_{m,\text{bits}} < \infty.$$

A proof of this statement would not by itself prove the Safe-Anchor Lemma, but it would establish that the required descendant-control information does not grow without bound across renormalization scales.

1.17 The corrected Bellman state

The original reduced Bellman state

$$V_m(y)$$

is numerically insufficient.

The fully resolved state

$$V_m(y, \Psi)$$

is unnecessarily expensive and potentially precision-dependent.

The revised candidate is

$$V_m^{(b)}(y, q), \quad q = Q_b(\xi_m(\pi)).$$

The finite-horizon recursion becomes schematically

$$V_m^{(L+1)}(y, q) = \inf_{\hat{\pi} \in \mathcal{A}_m^{(b)}(y, q)} \max \left\{ d_m(\hat{\pi}), V_{2m}^{(L)}(y', q') \right\},$$

where

$$(y', q') = \mathcal{T}_m^{(b)}(y, q; \hat{\pi})$$

is the canonical quantized child state.

The goal of Bellman search is only to discover a controlled invariant finite-resolution tube.

The theorem itself remains the one-step existential closure

$$\mathfrak{R}_m^{(b)} \longrightarrow \mathfrak{R}_{2m}^{(b)}.$$

1.18 Relation to the failed greedy policy

The earlier controlled line search showed

$$d_{\text{parent}} \uparrow \implies d_{\text{child}}^{\text{safe}} \downarrow$$

along a fixed parent fiber.

Thus the best present reduced profile need not produce the best future state.

The replicability intervention changes the objective once more.

The desired anchor is neither

$$\arg \min d_{\text{parent}}$$

nor necessarily

$$\arg \min d_{\text{child}}.$$

It is a representative satisfying simultaneously

safe now,
safe after doubling,
far from quantization boundaries,
deterministically reconstructible,
bounded inventory increment.

Thus some local geometric optimality may be intentionally sacrificed to obtain a reproducible hereditary orbit.

1.19 The revised resource tradeoff

The intervention exposes an explicit tradeoff:

$$\text{numerical precision} \longleftrightarrow \text{replicability margin}.$$

Exact phase tracking corresponds formally to

$$b = \infty.$$

The new program asks whether some finite

$$b < \infty$$

already suffices.

If a continuous candidate has descendant slack

$$\rho_0 + \eta$$

and the quantization error costs at most

$$L_{\xi} c_d 2^{-b},$$

then it is enough to choose

$$L_{\xi} c_d 2^{-b} \leq \eta.$$

Equivalently,

$$b \geq \log_2 \frac{L_{\xi} c_d}{\eta}.$$

Thus a uniform lower bound on slack and uniform upper bound on phase sensitivity imply a uniform finite bit depth.

This is the exact mathematical form of the accuracy–replicability tradeoff.

1.20 What the intervention has already resolved

The following conceptual problems are no longer formulated in their old form.

1. The objective is no longer to identify an exact phase fixed point.
2. The objective is no longer one-step contraction

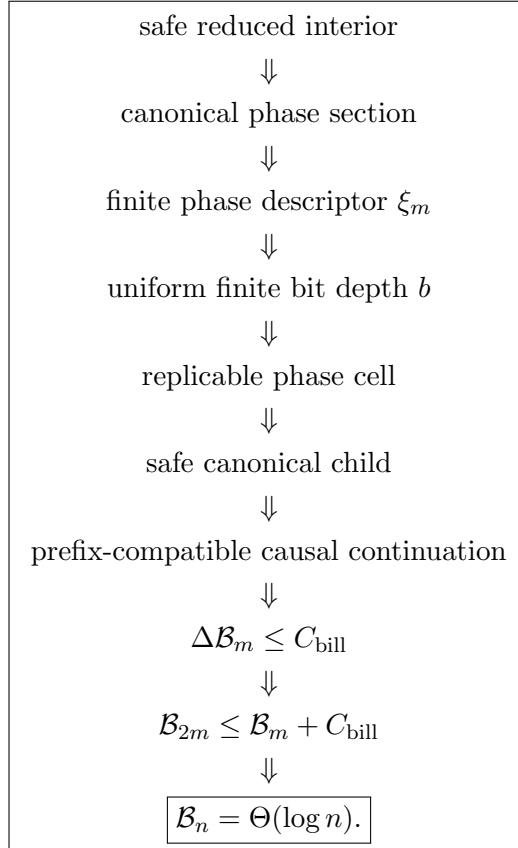
$$d_{2m}^{\text{safe}} \leq \lambda d_m^{\text{safe}}.$$

3. The objective is no longer greedy minimization of current target distance.
4. The reduced profile is no longer assumed to be a complete Markov state.
5. The growing m^2 -dimensional fresh phase is no longer required to be tracked with arbitrary precision.
6. A solver-dependent transport vertex is no longer regarded as a canonical continuation.

The remaining state is instead a robust finite-resolution phase class.

1.21 Current theorem ladder

The upper-bound proof has now been reduced to the following chain:



Every arrow is a separate theorem obligation.

1.22 What is proved

1. [PROVEN] The reduced Farkas margin is concave in the profile.

Hence any segment between two safe profiles remains safe at the corresponding interpolated margin.

2. [PROVEN] The compressed dual construction gives lower and upper LP bounds

$$\delta_{\text{lower}} \leq \delta_{\text{true}} \leq \delta_{\text{upper}}.$$

3. [PROVEN] The exact quartic renormalization law satisfies

$$\frac{\chi'}{\chi} = \frac{2}{rR^2}.$$

4. [PROVEN] At the critical fingerprint

$$(m, \mu_2, \mu_4) = \left(\frac{7}{9}, \frac{4}{3}, 6 \right),$$

one has

$$rR^2 = 2.$$

5. [PROVEN] The Common-Reserve Recycling Lemma reduces a depth- $2n$ continuation to one common reserve law valid for all depth- n leaves.

6. [PROVEN] A uniform bound

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + C$$

implies

$$\mathcal{B}_n = O(\log n).$$

7. [PROVEN] Together with the existing lower bound

$$\mathcal{B}_n = \Omega(\log n),$$

such a doubling estimate would yield

$$\mathcal{B}_n = \Theta(\log n).$$

1.23 What is numerically established

1. [NUMERICAL] One-step safe-deficit contraction is false on audited branches:

$$\frac{d_{513}}{d_{257}} \approx 2.882$$

and

$$\approx 1.176.$$

2. [NUMERICAL] On one fixed parent fiber,

$$d_{\text{parent}} \uparrow \implies d_{\text{child}}^{\text{safe}} \downarrow$$

with

$$R^2 \approx 0.9996.$$

3. [NUMERICAL] The same qualitative control signal appears on a genealogically distinct parent, but with substantially different strength.
4. [NUMERICAL] The reduced profile alone does not determine descendant quality.
5. [NUMERICAL] A noncanonical LP fiber selector gives unstable finite-difference descendant sensitivities.
6. [NUMERICAL] A canonical maximum-entropy section gives a stable finite-difference operator:

$$\varepsilon_J \approx 1.6 \times 10^{-3}.$$

7. [NUMERICAL] On the first canonical audit,

$$\sigma(J_{\text{eff}}) \approx (1.972, 1.008, 0.249),$$

with

$$R_{\text{bits}}^{\text{eff}} \approx 0.99.$$

8. [NUMERICAL] The measured shape residual is

$$\frac{\|R_{\text{shape}}\|_F}{\|J_{\text{raw}}\|_F} \approx 0.235.$$

1.24 What is open

1. [OPEN] Construct a finite-dimensional phase descriptor

$$\xi_m$$

that controls all descendant-safe directions relevant to the invariant tube.

2. [OPEN] Prove a scale-independent bound

$$d < \infty.$$

3. [OPEN] Prove a scale-independent quantization depth

$$b < \infty.$$

4. [OPEN] Prove a uniform phase-sensitivity bound sufficient to control quantization loss.

5. [OPEN] Prove a uniform quantization margin

$$\text{qmargin}_b \geq \gamma 2^{-b}.$$

6. [OPEN] Prove closure of the replicable invariant tube:

$$\mathfrak{K}_m^{(b)} \longrightarrow \mathfrak{K}_{2m}^{(b)}.$$

7. [OPEN] Lift reduced canonical viability to a full prefix-compatible causal tree.

8. [OPEN] Construct a common reserve with

$$\mathbb{E}R \leq C_{\text{bill}}$$

uniformly in m .

9. [OPEN] Prove the combined Replicable Safe-Anchor Lemma.

10. [OPEN]

$$\boxed{\mathcal{B}_n = \Theta(\log n)}.$$

1.25 Immediate falsifiable tests

The next numerical work should no longer optimize arbitrary exact phase points.

It should test the finite-information hypothesis directly.

For each audited state and scale:

1. choose one canonical interior section;
2. identify the leading descendant-sensitive phase coordinates;
3. quantize them at

$$b = 1, 2, 3, \dots;$$

4. reconstruct the canonical coupling from the quantized label;
5. measure

$$\delta_{\text{child}}, \quad d_{\text{child}}^{\text{safe}}, \quad \text{qmargin}, \quad \Delta\mathcal{B};$$

6. find the smallest b preserving the safe tube;
7. repeat across scales and genealogies.

The critical empirical sequence is therefore

$$b_{129}^*, \quad b_{257}^*, \quad b_{513}^*, \quad b_{1025}^*, \dots$$

The strongest possible numerical signal would be stabilization:

$$b_m^* \leq b_0 \quad \text{for all audited } m.$$

Failure is equally informative.

If

$$b_m^* \rightarrow \infty,$$

then the finite-phase formulation is false in its present form.

1.26 The revised QED bottleneck

Before the replicability intervention, the upper-bound problem appeared to require control of an increasingly complicated exact phase state.

After the intervention, the conjectural bottleneck is sharper:

$$\begin{aligned} &\exists d, b, \varepsilon_0, \rho_0, \gamma, C_{\text{bill}} < \infty \quad \text{independent of } m, \\ &\text{such that a deterministic canonical } d\text{-dimensional} \\ &\quad \text{phase descriptor quantized to } b \text{ bits} \\ &\quad \text{propagates the safe tube under } m \mapsto 2m \\ &\quad \text{with prefix-compatible inventory cost } \leq C_{\text{bill}}. \end{aligned}$$

If this statement is proved, then

$$\mathcal{B}_{2m} \leq \mathcal{B}_m + C_{\text{bill}},$$

hence

$$\mathcal{B}_n = O(\log n).$$

Combined with

$$\mathcal{B}_n = \Omega(\log n),$$

the desired result follows:

$$\mathcal{B}_n = \Theta(\log n).$$

1.27 Updated research principle

The intervention changes the research standard.

The target is no longer the most accurate available continuous representative.

The target is the least informative representative that still propagates the theorem.

Schematically,

continuous phase $\longrightarrow$ canonical phase $\longrightarrow$ quantized phase cell $\longrightarrow$ replicable descendant $\longrightarrow$ exact inventory

The principle may be summarized as

Canonicalize. Quantize. Propagate. Recycle. Verify.
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Canonicalization removes arbitrary fiber choices.

Quantization prevents precision from becoming a hidden resource.

Propagation establishes descendant viability.

Recycling converts viability into bounded inventory increment.

Verification remains exact.

1.28 Final checkpoint

The fixed-point ansatz failed.

The greedy contraction ansatz failed.

The reduced profile alone failed as a complete dynamic state.

Exact phase tracking is unnecessarily strong and numerically non-replicable.

The controlled invariant-tube mechanism remains viable.

The present candidate mechanism is now:

finite-resolution canonical phase dynamics + bounded common-reserve recycling.
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The remaining theorem is not that one privileged exact orbit converges.

It is that a finite amount of robust phase information suffices at every dyadic scale.

Equivalently:

the upper-bound problem has become a phase-bit sufficiency problem.

If phase-bit sufficiency and bounded inventory recycling are both proved uniformly, the asymptotic bill is closed:

$\mathcal{B}_n = \Theta(\log n).$
